

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Common to all Branches

Semester :II

Subject Code & Name: BTBS202P Engineering Physics

Max Marks: 60

Date: 20/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Magnetostriction method is used for the production of ultrasonic waves upto the frequency	REMEMBER (CO1)	1
	a. 200 KHZ b. 50 KHZ c. 100 KHZ d. 500 KHZ		
2	The oscillatory motion is said to be critically damped if -----	REMEMBER (CO1)	1
	a. $b^2 > \omega^2$ b. $b^2 < \omega^2$ c. $b^2 = \omega^2$ d. None of these		
3	In Newton's ring experiment the center appears to be _____ in reflected light.	REMEMBER (CO2)	1
	a. Bright b. Dark c. Mixed d. None of these		
4	The substance which rotates the plane of polarization to left is called as	REMEMBER (CO2)	1
	a. Dextrorotatory b. Levorotatory c. Oscillatory d. None of these		
5	Wavelength of He-Ne laser is	REMEMBER (CO2)	1
	a. 6943 A ⁰ b. 6428 A ⁰ c. 6328 A ⁰ d. 6398 A ⁰		
6	Optical fiber operates on the principle of	REMEMBER (CO2)	1
	a. Total internal reflection b. Total external reflection c. Total internal refraction d. Total external refraction		

7	Dual nature [particle and wave] of matter was proposed by				REMEMBER (CO3)	1
	a. de Broglie	b. Planck	c. Newton	d. Heisenberg		
8	The mixture filled in G.M. tube is ,				REMEMBER (CO3)	1
	a. Argon + Ethyl Alcohol	b. Neon+ Ethyl Alcohol	c. Argon+ Methyl Alcohol	d. Neon+ Methyl Alcohol		
9	Number of atoms per unit cell for Body centered Cubic structure is				REMEMBER (CO4)	1
	a.2	b. 6	c. 4	d. 12		
10	Coordination Number for Face centered Cubic structure is				REMEMBER (CO4)	1
	a.6	b. 8	c. 12	d. 10		
11	In superconductor the value of susceptibility is,				REMEMBER (CO)	1
	a.0	b. 1	c. -1	d. 2		
12	Hall effect is associated with,				REMEMBER (CO)	1
	a. Conductors	b. Semiconductors	c. Both a and b	d. None of these		
Q. 2	Solve the following.					12
A)	Define free vibrations. Derive Differential Equation for free vibrations.				REMEMBER & UNDERSTAND (CO1)	6
B)	State magnetostriction effect. Explain magnetostriction method for the production of ultrasonic waves.				REMEMBER & UNDERSTAND (CO1)	6
Q.3	Solve the following.					12
A)	Explain the construction and working of Ruby laser.				UNDERSTAND (CO2)	6
B)	Derive an expression for numerical aperture of optical fiber.				UNDERSTAND (CO2)	6

Q. 4	Solve Any Two of the following.		12
A)	Explain the construction & working of Bainbridge mass spectrograph with labeled diagram.	UNDERSTAND (C03)	6
B)	Derive Schrodinger's time independent wave equation.	UNDERSTAND (C03)	6
C)	Explain construction and working of Geiger Muller counter.	UNDERSTAND (C03)	6
Q.5	Solve Any Two of the following.		12
A)	Define and find atomic radii for Simple cubic, body centered cubic and face centered cubic structure.	UNDERSTAND (C04)	6
B)	Explain continuous and characteristics X-ray spectra.	UNDERSTAND (C04)	6
C)	Derive the relation between lattice parameter 'a' and crystal density 'ρ'. Copper has FCC structure and its atomic radius is 1.278 Å. Calculate density of Cu. Given atomic weight of Cu=63.5.	UNDERSTAND (C04)	6
Q. 6	Solve Any Two of the following.		12
A)	Explain Diamagnetic, Paramagnetic and Ferromagnetic materials with examples and diagram.	UNDERSTAND (C0)	6
B)	State & explain Meissner effect for superconductors.	UNDERSTAND (C0)	6
C)	Explain Conductor, Semiconductor and Insulator on the basis of band theory of Solids.	UNDERSTAND (C0)	6
	*** End ***		